

Alex Merlo

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CAREER FOCUS: Software Engineering Intern

As a computer science student at Oakland University, I am seeking an internship to improve my skills as a developer. I bring a strong foundation in programming, hands-on experience with data analytics, and the ability to collaborate effectively in team environments.

Education:

Bachelor of Science, Computer Science; 3.38 GPA

Sep 2021 - Apr 2026

Oakland University, Rochester, MI

Course Work:

- Data Structures and Algorithms
- Discrete Mathematics
- Linear Algebra
- Design & Analysis of Algorithms

Experience:

Data Analyst CO-OP

May 2024 - Sep 2024

Hutchinson, Auburn Hills, MI

- Used QlikSense to create interactive dashboards for the manufacturing plants to track their monthly KPIs
- Managed data loading to enable effective data exploration and analysis
- Utilized Qlik's Data Manager to link databases and Excel sheets to the visualizations

Engineering CO-OP

May 2023 - Jan 2024

Hutchinson, Auburn Hills, MI

- Used CAD to find key dimensions and identify any problems that would come up with parts
- Created presentations to be presented to OEMs that would be used to inform them of the estimated material used and changes that were made from the original design
- Collaborated with others to ensure design feasibility and adherence to client specifications.

Projects:

Fit Pal Pro

- A react application that allows users to track their workouts, become friends with other people, and view their friend's recent workouts.
- Designed SQL queries to retrieve user data and handle dynamic updates.
- Created and implemented secure GET and POST requests for workout tracking and friendship management.
- Incorporated password hashing to ensure data security

Quadratic Surface Visualizer

- Creates a Quadratic Surface class in Python that analyzes and visualizes quadratic surfaces in 3D space using QR factorization, eigenvalues, and shape classification while aligning it with a more observable axis

Technical Skills:

Programming Languages: Python, Java, Javascript, HTML, CSS, C

Technologies: Pygame, JavaFX, VsCode, QlikSense, Github, Flask

Skills: Data Structures & Algorithms, Data Visualization, Communication, Excel

Activities:

- OU Competitive Programming - Member
- Computer Building - Sourced parts and assembled a PC